ABDELRAHMAN ELEWAH

abdelrahman.elewah@gmail.com

+1 289-939-6665

PROFILE

- 7+ years as a Research/Teaching Assistant at Benha University and Ontario Tech University. We designed the SensorsConnect framework and participated in the developing data visualization method (RadViz-plotly).
- 3+ years Teaching Computer Architecture, Micro Processors, and Embedded Systems courses and leading FPGA/VHDL tutorials.
- Design a multi-modulation for low earth orbit satellite based on software-defined radio.
- Participating as a Research Associate in Ontario Tech University's Regional Future Workforce project team that bridges the gap between industry and academia to equip students with skills required for the mobility sector.
- 1 year experience working in 2U as a TA for Data Analytics Boot Camp.
- 3+ years of leadership at Tamra IoT. Tamra IoT won a seed fund in the InnoEgypt program at the ICT track.
- Hands-on experience designing IoT systems and projects that combine embedded systems, cloud functions, and databases.
- Design the AMA system that integrates the elevator with smart tablets and features. The project was rewarded 1st in the Mobile Application Launchpad (MAL) competition.

PROFESSIONAL EXPERIENCE

Teaching Assistant, 2U, Toronto, Canada 2023– present

- Contribute to the success of an online Data Analytics Boot Camp specializing in training data professionals.
- Assist learners in acquiring essential practical and technical skills for modern data analysis.
- Guide learners through real-world projects tailored to industry sectors like Finance, HR, Healthcare, and Government**.**

Research Assistant, Ontario Tech University, Dept Electrical, Computer, and Software Eng., Ontario, Canada 2020– present

- Design a framework called SensorsConnect: a search engine for real-time sensing data.
- Designed the ThingsDriver standard to manage/control IoT nodes from the cloud and published it as a paper.
- Collaborate in a project funded by OVIN that focuses on developing curriculums for autonomous vehicles and conducting research on fields such as autonomous vehicles, software-defined vehicles (SDV), and digital twins.
- Design a data visualization method, RadViz-Plotly, to visualize high-dimensional data.
- Collaborate in an Aircraft Deceleration Early Warning system.
- Introduced a data visualization method, RadViz-Plotly, to visualize high-dimensional data and published it as a paper.

Teaching Assistant, Ontario Tech University, Dept Electrical, Computer, and Software Eng., Ontario, Canada 2020 – present

- Teaching/Leading tutorials in Electrical/Software Engineering courses:
  - Digital systems
  - Circuit Analysis
  - Systems programming
  - Embedded Systems
  - Intro to Programming
  - Data Management Systems

Co-Founder, Tamra-IoT 2018- present

- Collaborate in designing Tamra IoT platform for Makers and STEM students.
- Collaborate in managing the business.
- Maintain ThingsDriver, an open-source package used to manage/control Tamra nodes from the cloud or Tamra mobile app.

Research Assistant, Benha University, Dept. of Electrical Engineering, Benha, Egypt 2014– 2019

- Designed multi-modem based on Software Defined Radio (SDR),
- Introduced an algorithm to estimate the signal-to-noise ratio and published it as a paper.

Teaching Assistant, Benha University, Dept. of Electrical Engineering, Benha, Egypt 2014 – 2019

- Teaching/Leading tutorial in Electrical Engineering courses
  - Computer Programming - Computer Architecture
  - Micro-processor-based systems - Electrical and Electronics applications
  - Field theory

Part-time Teaching Assistant, Arab Academy for Science, Technology & Maritime Transport-Smart Village, Giza, Egypt 2019

- Teaching/Leading tutorials in Computer Science courses
  - Digital Logic - Computer Networks

EDUCATION

Master of Science in Electrical Engineering, Benha University, Benha, Egypt 2018

- **Thesis:** Multi-modulation Low Earth Orbit Satellite Based on Software Defined Radio
- Implement and test Mth Root Mth Power SNR MPSK Estimator using USRP kit.

Bachelor of Electrical Engineering, Benha University, Benha, Egypt 2012

List of Publications

[1] Elgazzar, K., Khalil, H., Alghamdi, T., Badr, A., Abdelkader, G., Elewah, A., & Buyya, R. (2022). Revisiting the internet of things: New trends, opportunities and grand challenges. Frontiers in the Internet of Things, 1, 1073780.

[2] A. Elewah, W. M. Ibrahim and K. Elgazzar, "Paving the Way for Massive Public Sensing as a Service," 2022 IEEE 47th Conference on Local Computer Networks (LCN), 2022, pp. 391-394, doi: 10.1109/LCN53696.2022.9843516.

[3] A. Elewah, W. M. Ibrahim, A. Rafıkl and K. Elgazzar, "ThingsDriver: A Unified Interoperable Driver for IoT Nodes," 2022 International Wireless Communications and Mobile Computing (IWCMC), 2022, pp. 877-882, doi: 10.1109/IWCMC55113.2022.9824989.

[4] A. Elewah, A. A. Badawi, H. Khalil, S. Rahnamayan and K. Elgazzar, "3D-RadViz: Three Dimensional Radial Visualization for Large-Scale Data Visualization," 2021 IEEE Congress on Evolutionary Computation (CEC), 2021, pp. 1037-1046, doi: 10.1109/CEC45853.2021.9504983.

[5] A. Elewah, A Taman, AA Awamry, M Elbahy "Mth Root Mth Power SNR MPSK Estimator", 12th International Conference Mechatronics, Brno, Czech Republic 6-8/09/2017.

ADDITIONAL INFORMATION

- Advanced Level Computer Skills: Programming with C, C++, Python, Javascript, VHDL, Microsoft Office
- Start gaining experience in Docker, Containers, and Dev containers Technologies.
- Familiar with development environments like VScode, PlatformIO, STM32Cube, Proteus, Arduino
- Data Management experience: SQL, NoSQL: MYSQL, MongoDB, Postgres, SQLite.
- A Volunteer at Enactus Ontario Tech.
- Participated in organizing Egyptian Engineering Day for 4 years.

COURSES

All Certificates Provided Upon Request

AWS Machine Learning Foundation 2022, Udacity 2022

- The course covers three topics:
  - Fundamentals of supervised and unsupervised learning ML, including the process steps of solving ML problems
  - Machine Learning with Amazon Web Services (AWS) covers some advanced topics: computer vision, reinforcement learning, and generative AI, in addition to using AWS ML tools.
  - Software Engineering Practices covers writing clean and modular code, code efficiency, refactoring, documentation, version control, testing code, logging, and conducting code review.

Data Modeling, Data camp, MongoDB 2022

- The course covers:
  - How to create data models for MongoDB.
  - Deal with a few techniques, from a straightforward process for simple schemas to more complex ones for large teams and projects.
  - Frequent patterns to apply when modelling and will be able to apply them in your designs.

A List of Google Cloud Courses, Coursera 2020

- How Google Does Machine Learning: covers the best practices for implementing machine learning on Google Cloud.
- Google Cloud Platform Fundamentals: covers essential concepts and terminology for working with Google Cloud.
- Industrial IoT on Google Cloud: presents Industrial Internet of Things on Google Cloud.

A List of Machine Learning courses, Data camp 2020

- Introduction to Data Visualization in Python
- Introduction to Deep Learning in Python
- Manipulating Time Series Data in Python
- Model Validation in Python
- Cleaning Data in Python
- ARIMA Models in Python
- Dealing with Missing Data in Python
- Dimensionality Reduction in Python
- Feature Engineering for Machine Learning in Python
- Visualizing Time Series Data in Python
- Analyzing IoT Data in Python